

DATA ANALYSIS INTERNSHIP

MediCare+ Hospital Analytics

Data Quality, Measures and Decision Support on 247 Admissions

Data Cleaning • Measures and KPIs • Statistical Tests • Insights • Interactive Dashboard

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1 Executive summary

The brief asks for a complete cleaning, the relevant measures and KPIs, an interactive dashboard and clear insights, along the chain *raw data, clean data, measures, insights, decision support*. All are delivered, together with the notebook that reproduces every number and figure of this report.

The register holds **247 admission episodes** from 75 distinct patients, attended by 7 physicians across 10 diagnoses, over the calendar year 2023.

The primary finding is financial. The billing column sums to **618,995**. That figure cannot be used. Of the 247 episodes, **195 carry one of two placeholder amounts**, 999 or 3,852. Only 52 episodes hold a genuine amount, and the revenue they support is **153,155**. The raw total overstates validated revenue by a factor of four.

Validated revenue is a floor, not an estimate. The 52 genuine bills sit almost entirely on short stays: none of the 43 stays longer than 12 days has one. Because the bill rises with the stay, the mean validated bill of 2,945 understates the typical bill and cannot be scaled up to the 247 episodes.

The second defect breaks the operational benchmark. 29 episodes, 11.7% of the file, record a discharge before the admission. Length of stay is computed on the 218 valid stays: mean **6.74 days**, median 4 days.

Length of stay, not severity, drives the bill. The bill rises by about 530 per day of stay (Spearman $\rho = 0.62$, $p < 0.001$). Severity explains neither the stay ($p = 0.84$) nor the bill ($p = 0.28$), and the severity mix does not differ between physicians ($p = 0.34$).

The monthly profile is compatible with chance. Admissions do not differ significantly across months ($p = 0.17$); a seasonal staffing roster built on this single year would rest on noise.

What the register supports well is volume, case mix, acuity and cover. Those are the numbers a capacity decision can rest on today.

2 Data and method

2.1 The register

One row is one admission episode. The columns used are the episode identifier, the patient's name, age and gender, the diagnosis, the severity grade (Low, Medium, High), the attending physician, the admission and discharge dates, the billed amount and the insurance cover (Private, Government, Uninsured). The period runs from 1 January to 28 December 2023.

2.2 Process

The analysis follows five steps, each implemented as a section of the notebook `MediCare-Hospital-Analytics.ipynb`:

1. **Quality audit.** Four routine checks (identifiers, completeness, category labels, ranges), then two targeted checks: repetition of a billed amount and chronology of the dates.
2. **Cleaning.** Every defect is flagged; no row is deleted.
3. **Measures.** Each indicator is computed on the population that can support it, and that population is stated with the value.
4. **Tests.** Every segment difference is tested before it is reported: Kruskal–Wallis for skewed measures across groups, chi-square for mixes and monthly counts, Spearman rank correlation for associations, Mann–Whitney for the representativeness of the genuine bills.
5. **Decision support.** Each finding is translated into an action and an owner, and exposed in an interactive dashboard.

The analysis uses Python only (pandas, NumPy, SciPy, Matplotlib). No spreadsheet and no business-intelligence tool is used at any stage.

3 Data quality

Table 1: Checks performed on the register as supplied

Check	Result
Admission episodes	247
Duplicate episode identifiers	0
Missing values in core columns	0
Spelling variants in category labels	0
Distinct patients	75
Attending physicians	7
Distinct diagnoses	10
Episodes with a placeholder bill (999 or 3,852)	195 (78.9%)
Episodes with a discharge before admission	29 (11.7%)
Episodes usable for revenue	52 (21.1%)
Episodes usable for length of stay	218 (88.3%)

The four routine checks find nothing. A report that stopped there would declare the file clean.

3.1 Why no routine check finds the billing defect

999 and 3,852 are both legal amounts inside the observed range. The cells are populated, so a completeness audit sees nothing; the values fall within bounds, so a validity rule sees nothing; the column type is correct, so a schema check sees nothing. Only the repetition gives the defect away. Among the other 52 amounts, no value occurs twice; 999 occurs 100 times and 3,852 occurs 95 times. An amount repeated on 40% of episodes is not a charge, it is a system default.

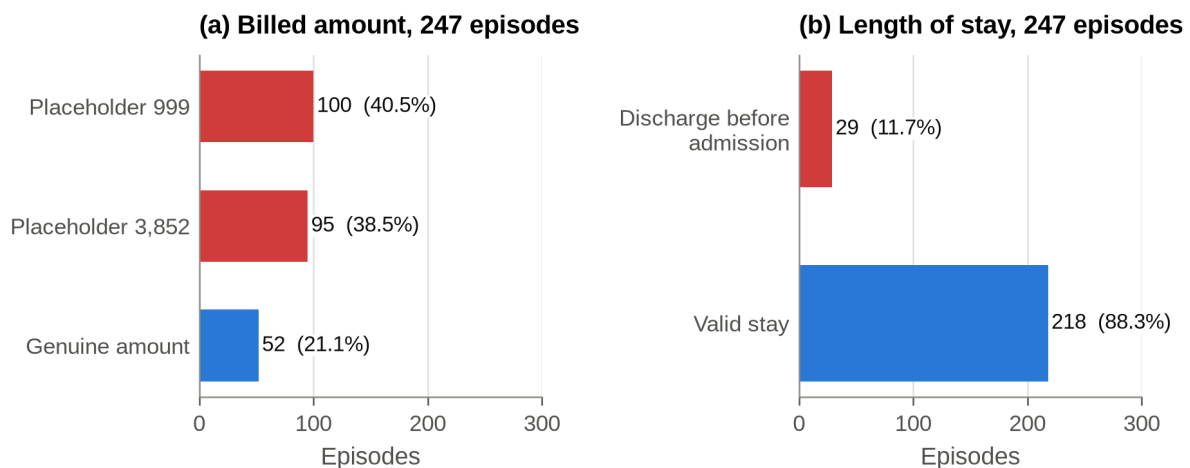


Figure 1: Episodes by billing status (a) and by stay status (b). Defects are shown in red. Only 52 of the 247 bills are genuine, and 29 stays end before they begin.

The flags recomputed from the primary columns agree exactly with the helper columns supplied with the file (`BillIsPlaceholder`, `DateError`, `LengthOfStay`), so both defects are confirmed by two independent routes.

4 Cleaning

Table 2: One decision per problem, with the reason for it

Problem	Count	Decision and reason
Placeholder bills	195	Flagged, not deleted. The rows carry a valid diagnosis, physician, severity and cover; deleting them would discard the clinical record to repair one column. <code>Bill_clean</code> holds the amount only where it is genuine.
Impossible stays	29	<code>LOS_valid</code> left empty, not set to zero. Zero denotes a same-day discharge, a real clinical event; writing it would invent 29 of them.
Insurance labels	3	Mapped to Private, Government and Uninsured, as stable keys for the dashboard filters.
Duplicates	0	None removed: identifiers are unique.

Every affected row keeps its flag, so a reader can include or exclude it and see the difference.

No row was deleted: the clean file has the same 247 rows as the raw file.

5 Indicators

Table 3: Each indicator with the population it is computed on

Indicator	Value	Computed on
Admission episodes	247	all episodes
Distinct patients	75	all episodes
Attending physicians	7	all episodes
Distinct diagnoses	10	all episodes
Mean patient age	47.4 years	all episodes
Female share	52.6%	all episodes
High severity share	36.0%	all episodes
Uninsured share	29.1%	all episodes
Mean length of stay	6.74 days	218 valid stays
Median length of stay	4 days	218 valid stays
Validated revenue	153,155	52 genuine bills
Mean validated bill	2,945	52 genuine bills
Median validated bill	2,704	52 genuine bills
Raw bill column sum	618,995	not for decisions

Three indicators name their own base. Revenue and the mean bill rest on 52 episodes; length of stay rests on 218. Naming the base beside the number is better than a footnote, because a footnote does not travel when a figure is copied into a slide.

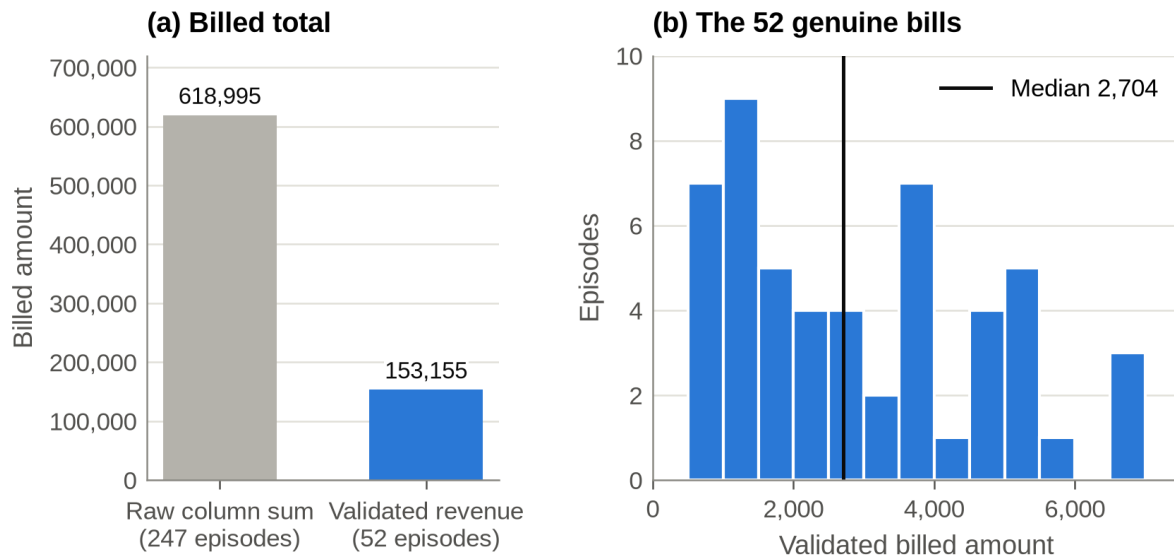


Figure 2: Raw bill column sum against validated revenue (a), and distribution of the 52 genuine bills in bins of 500 (b). 465,840 of the raw total rests on placeholder amounts. The genuine bills range from 503 to 6,763, with a median of 2,704.

6 Results

6.1 Case mix

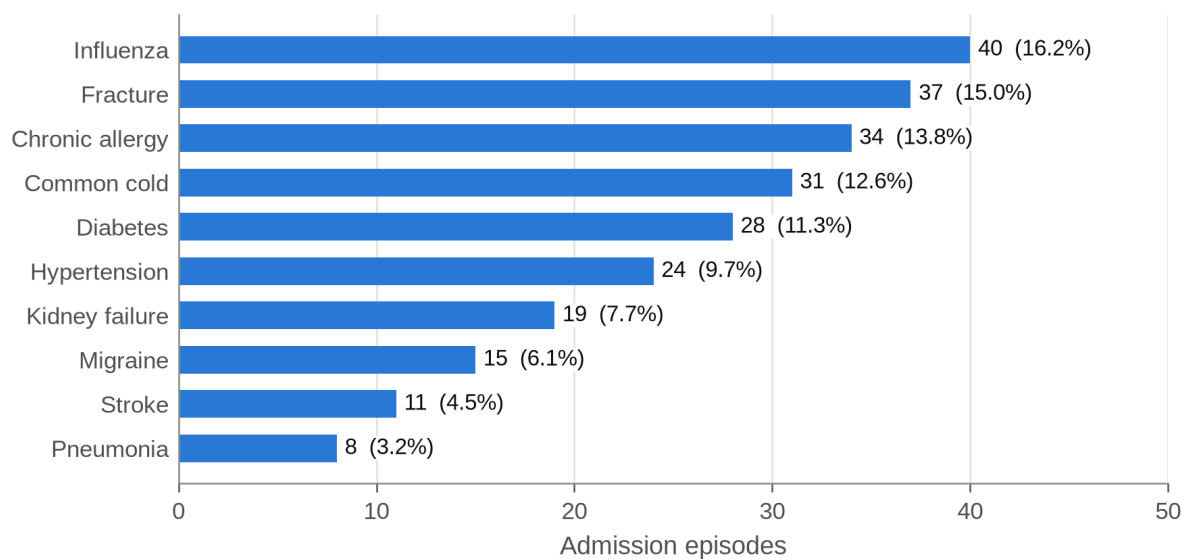


Figure 3: Admission episodes by diagnosis. Influenza (40), fracture (37) and chronic allergy (34) account for 45% of admissions.

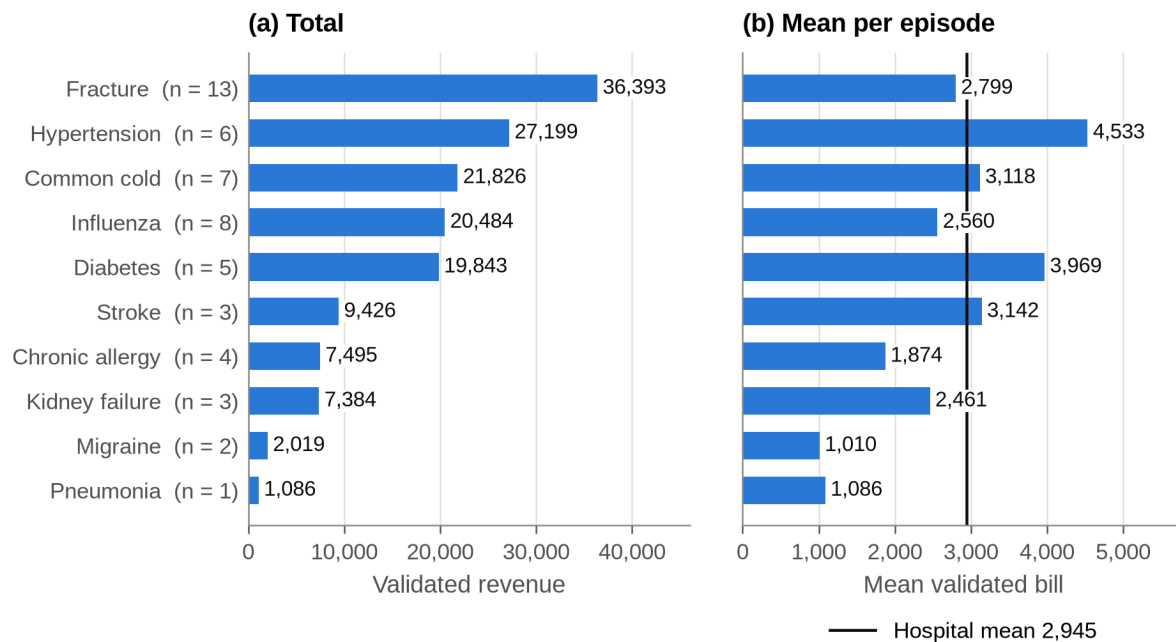


Figure 4: Validated revenue by diagnosis: total (a) and mean per episode (b), in the same diagnosis order. The number of genuine bills is given beside each diagnosis. Fracture leads total revenue on 13 bills; the highest means, hypertension (4,533) and diabetes (3,969), rest on 6 and 5 bills and are indicative only.

6.2 Activity through the year

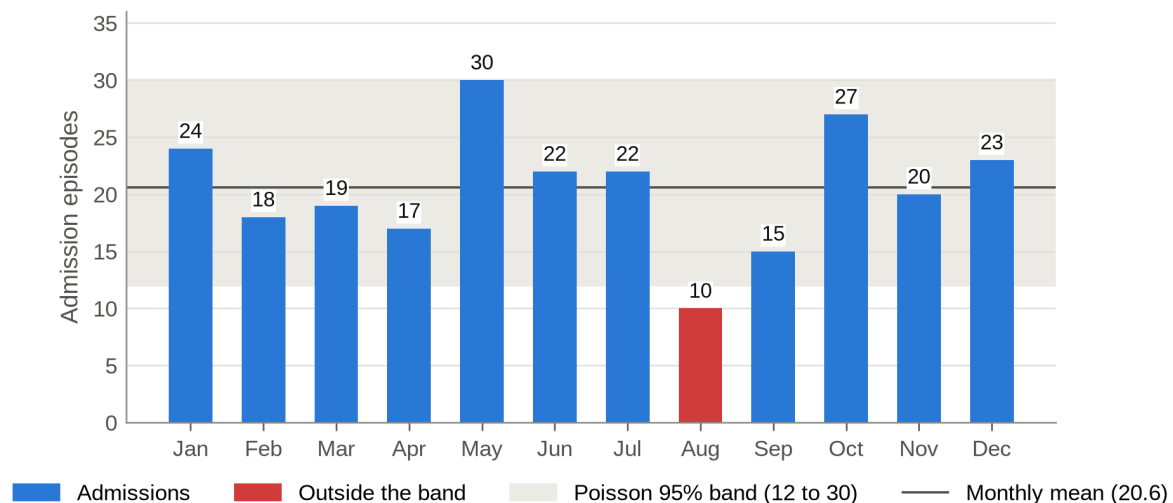


Figure 5: Monthly admissions against the central 95% range of a Poisson count with the observed mean (20.6 per month). Only August (10) falls outside the band, about what chance alone produces over twelve months (0.6 expected). May (30) sits on the upper edge of the band. A chi-square test does not reject an even spread across months ($p = 0.17$).

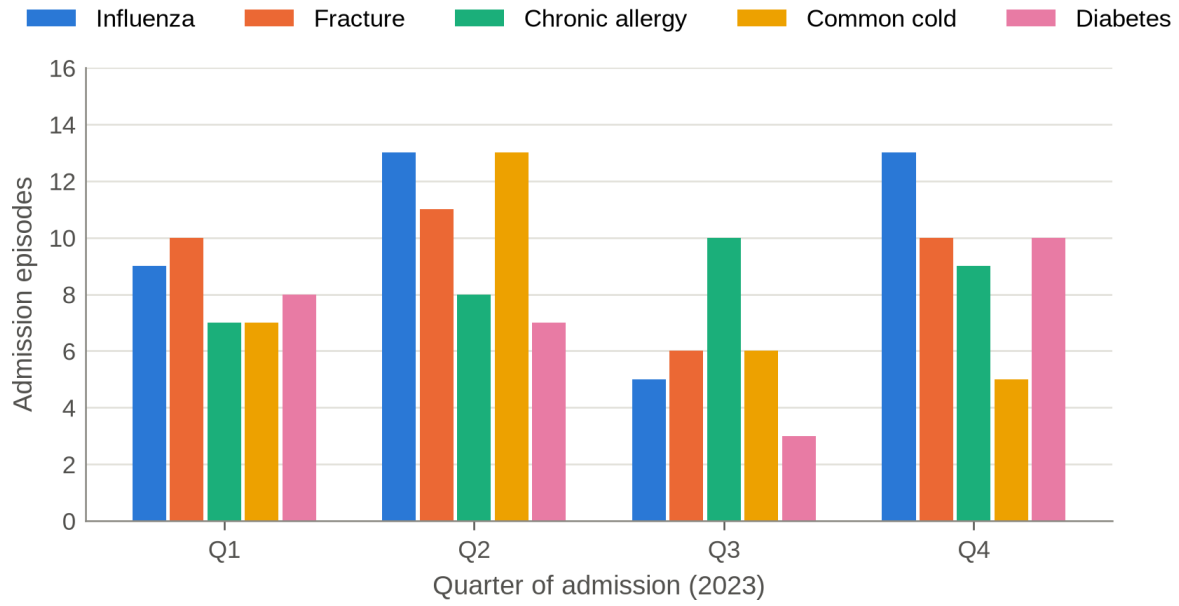


Figure 6: Quarterly admissions for the five leading diagnoses. The counts are given in Table 4.

Table 4: Admissions by quarter for the five leading diagnoses

Diagnosis	Q1	Q2	Q3	Q4	Year
Influenza	9	13	5	13	40
Fracture	10	11	6	10	37
Chronic allergy	7	8	10	9	34
Common cold	7	13	6	5	31
Diabetes	8	7	3	10	28
All diagnoses	61	69	47	70	247

6.3 Acuity, cover and physicians

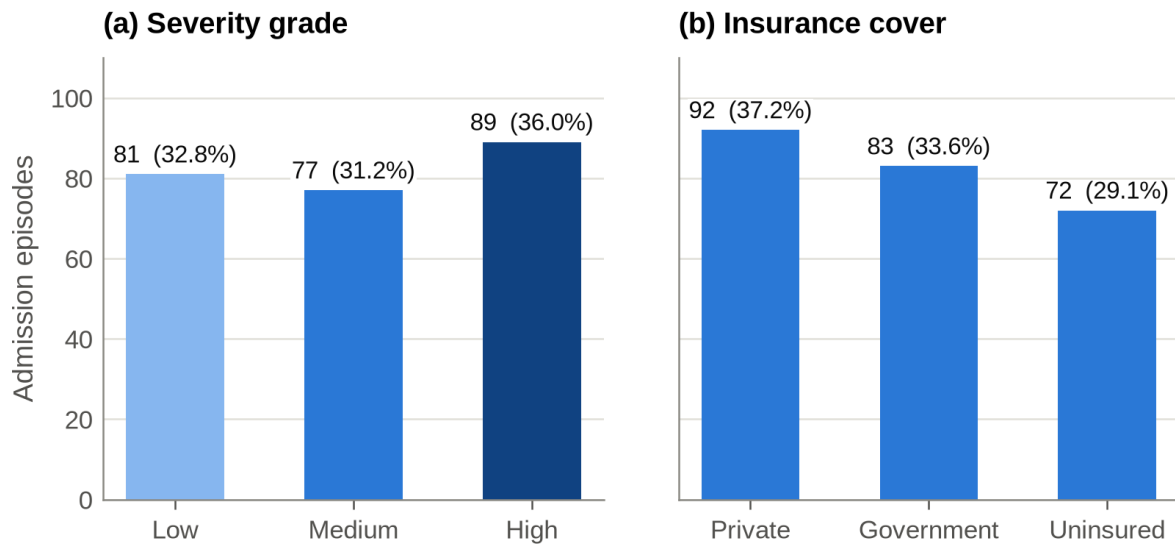


Figure 7: Admissions by severity grade (a) and by insurance cover (b). 36.0% of admissions are graded high severity and 29.1% arrive uninsured.

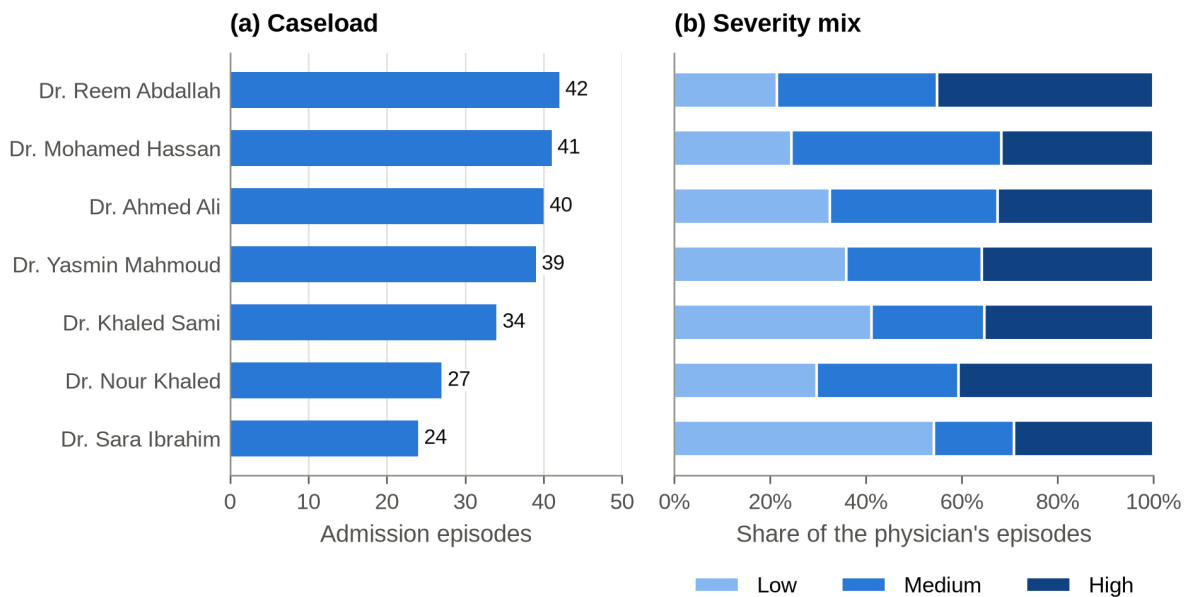


Figure 8: Caseload by physician (a) and severity mix of each caseload (b). Caseloads range from 24 to 42 episodes. The severity mix does not differ significantly between physicians ($\chi^2 = 13.4$, 12 df, $p = 0.34$): a ranking on this sample would judge noise.

6.4 Stay, bill and what drives them

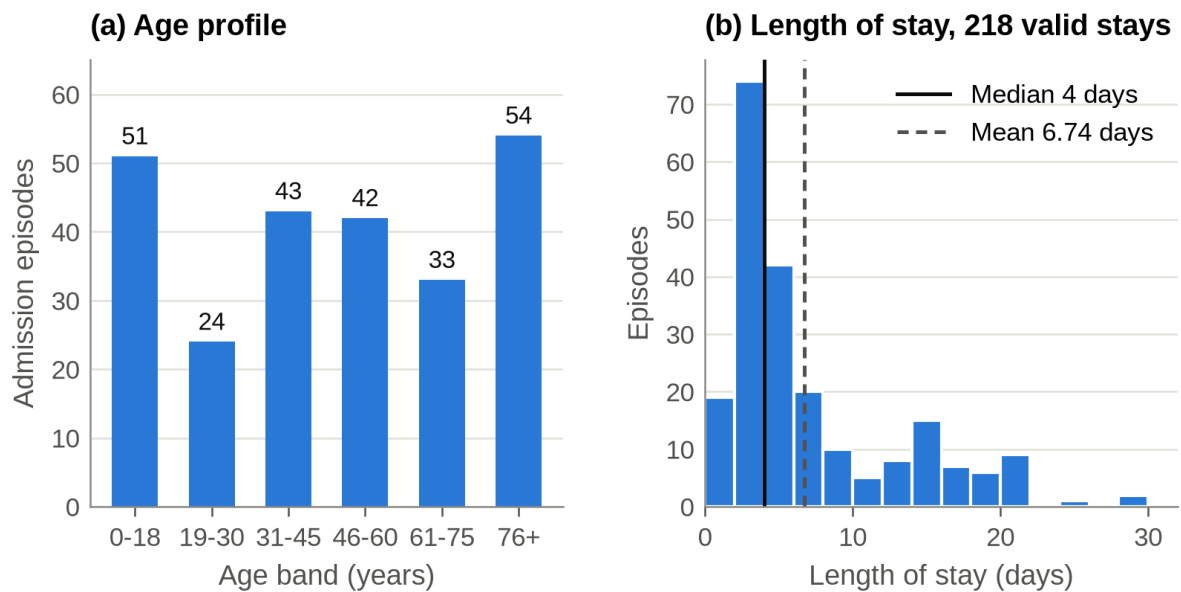


Figure 9: Admissions by age band (a) and distribution of the 218 valid stays in bins of two days (b). The stay distribution is right-skewed: the mean (6.74 days) exceeds the median (4 days), and 24% of stays exceed 9 days.

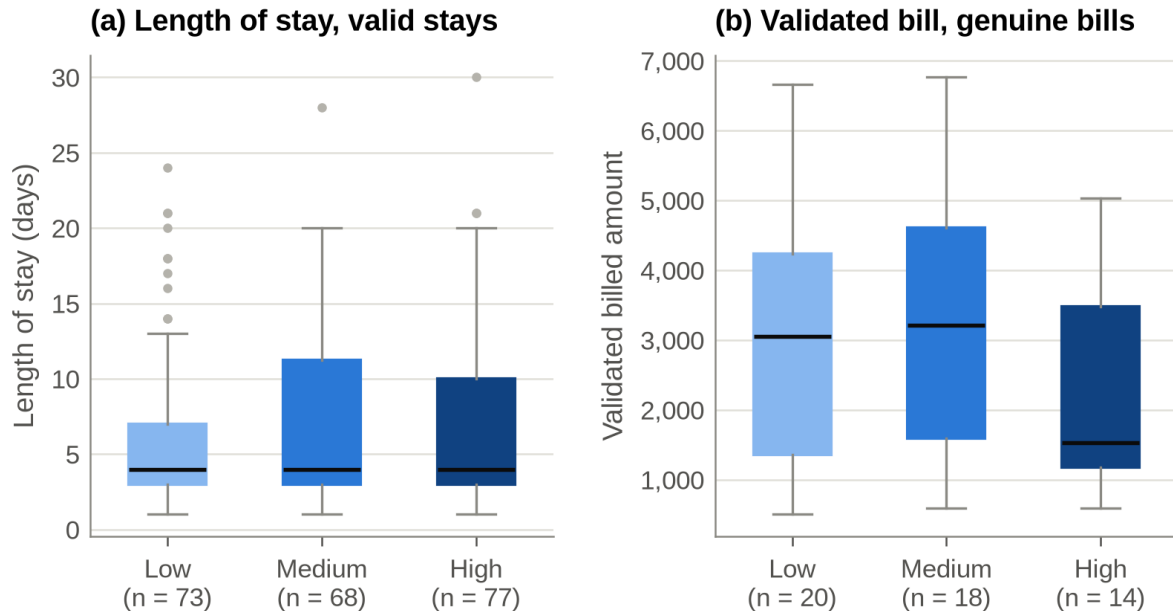


Figure 10: Length of stay (a) and validated bill (b) by severity grade. Boxes show the interquartile range, black lines the median, whiskers 1.5 times the interquartile range. The median stay is 4 days in all three grades (Kruskal–Wallis $p = 0.84$); the bill does not differ either ($p = 0.28$).

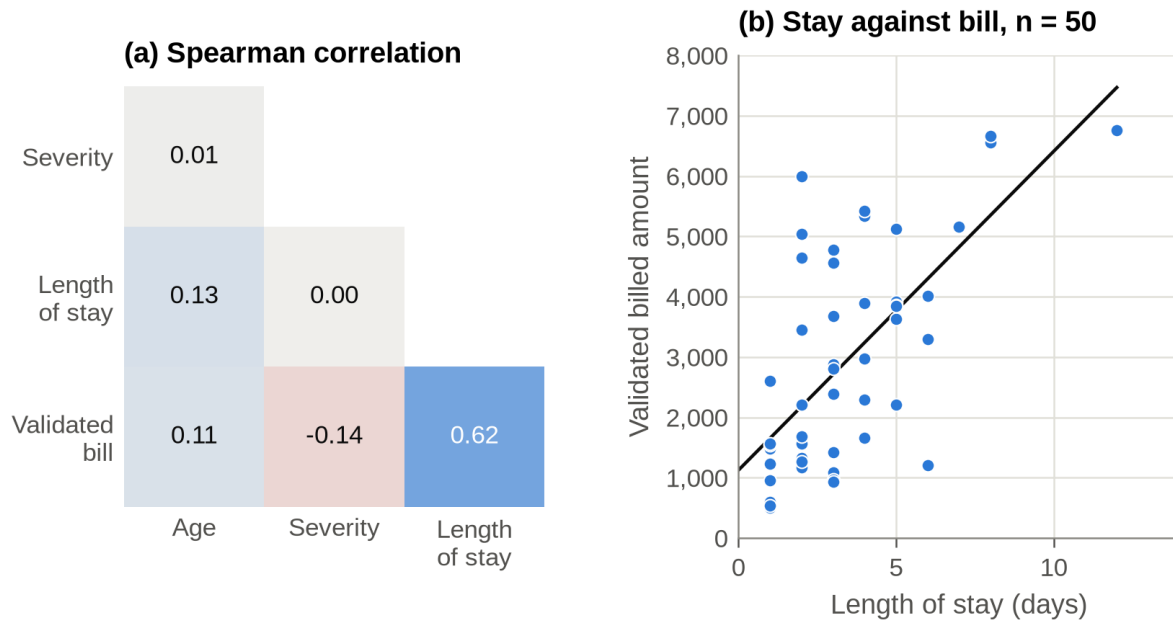


Figure 11: Spearman rank correlations between the four numeric measures, each on the episodes where both values are valid (a), and validated bill against length of stay for the 50 episodes with both (b). The line is the least-squares fit, $\text{bill} = 1,123 + 529 \times \text{days}$. Length of stay is the only measure associated with the bill ($\rho = 0.62$, $p < 0.001$).

6.5 Representativeness of the genuine bills

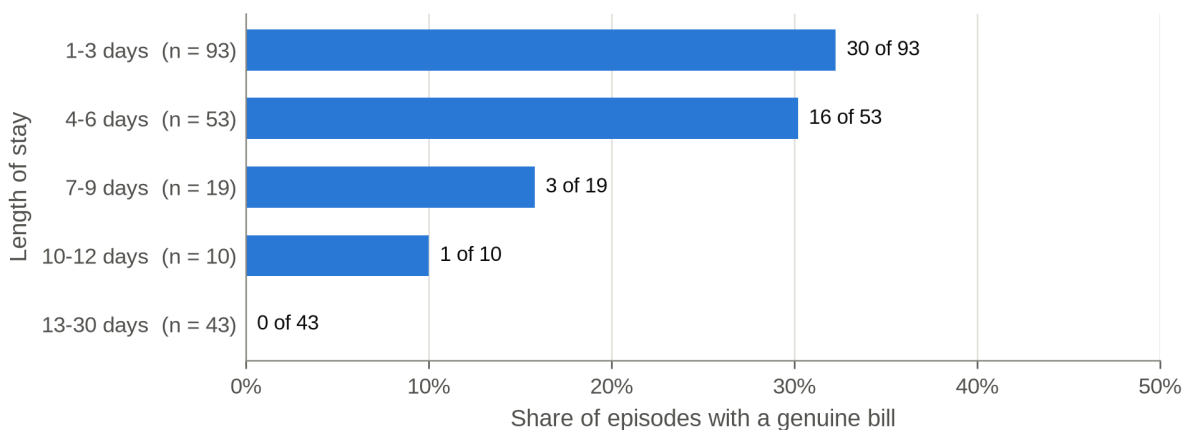


Figure 12: Share of episodes with a genuine bill, by length-of-stay band. The share falls from 32% for stays of 1 to 3 days to 0% for the 43 stays of 13 days or more. Genuine bills sit on shorter stays than placeholder bills (median 3 against 5 days, Mann–Whitney $p < 0.001$).

This result governs how every financial figure in this report must be read. The 52 genuine bills are not a random sample of the 247 episodes: they come from short stays, and the bill rises with the stay. Their mean therefore understates the typical bill, and multiplying it by 247 would not estimate annual revenue.

7 Statistical tests

Table 5: Tests behind every segment statement of this report

Question	Test	Statistic	p	n
Stay differs by severity	Kruskal–Wallis H	0.35	0.841	218
Bill differs by severity	Kruskal–Wallis H	2.51	0.285	52
Severity mix differs by physician	χ^2 (12 df)	13.40	0.341	247
Severity mix differs by cover	χ^2 (4 df)	1.19	0.880	247
Admissions differ by month	χ^2 (11 df)	15.40	0.165	247
Stay associated with bill	Spearman ρ	0.62	< 0.001	50
Genuine bills on shorter stays	Mann–Whitney U	2,458	< 0.001	218

Two results are significant at the 5% level, and both concern the bill: it follows the length of stay, and the genuine bills over-represent short stays. None of the segment differences by severity, physician, cover or month is significant.

8 Insights and recommendations

8.1 Billing integrity is the primary risk

195 of 247 episodes carry a placeholder amount. Validated revenue is 153,155 against a raw column sum of 618,995. The 52 genuine bills also over-represent short stays, so validated revenue is a floor rather than an estimate.

Recommendation. Publish finance indicators on validated bills only, state the 52-episode base beside every amount, and do not extrapolate them. Audit the charge-capture path that writes 999 and 3,852 before the next reporting cycle, starting with long stays, where no genuine bill exists. *Owner: finance and billing IT.*

8.2 Date errors break the stay benchmark

29 episodes record an impossible stay. Mean length of stay on the 218 valid stays is 6.74 days, median 4 days.

Recommendation. Block the save when the discharge date precedes the admission date. One validation rule at entry would have preserved 11.7% of the year’s operational record. *Owner: admission system IT.*

8.3 Length of stay, not severity, drives the bill

The bill rises by about 530 per day of stay over the 1 to 12 days observed. Severity predicts neither the stay nor the bill, and medium-severity stays are as long as high-severity ones.

Recommendation. Manage cost through discharge planning for long stays: one valid stay in four exceeds 9 days. Audit how the severity grade is assigned at admission; a triage grade that predicts neither stay nor bill needs correcting. *Owner: medical direction.*

8.4 Case mix is concentrated; acuity and cover create pressure together

Influenza, fracture and chronic allergy account for 45% of admissions. 36.0% of admissions are graded high severity and 29.1% arrive uninsured.

Recommendation. Align bed, pharmacy and pathway capacity with the three leading diagnoses. Move financial counselling for uninsured patients to admission rather than discharge. *Owner: operations and patient services.*

8.5 The monthly profile does not support a seasonal roster

Admissions are compatible with an even spread across the year ($p = 0.17$). Only August falls outside the range chance produces.

Recommendation. Do not build a staffing roster on the shape of a single year. Re-test seasonality when a second year is available. *Owner: operations.*

9 The interactive dashboard

`dashboard.html` is a single self-contained file: no server and no installation. It is built with Python and Plotly. The left navigation bar opens seven views: Overview, Dashboard, Patients, Clinical, Financial, Data Quality and Insights. Key indicators, charts and tables are rendered in the browser, and filters on month, physician, diagnosis, insurance, gender and severity update every visual at once. A live-data dock reloads a cleaned CSV or Excel file and recomputes the indicators, charts, correlations and insights.

An optional WebSocket server, `websocket_server.py`, streams the indicators from the clean register to the dashboard and follows its filters (`WEBSOCKET.md` documents the protocol). It computes revenue on genuine bills and length of stay on valid stays, so a streamed figure always matches this report. The dashboard works without it.

9.1 Reading the dashboard against this report

Three dashboard elements use a different base from the report. They are listed here so that no reader meets an unexplained difference.

Table 6: Reconciliation between dashboard elements and report indicators

Dashboard element	Base used and report equivalent
<i>Revenue</i> tile (618,995); <i>Revenue by insurance</i> , <i>Revenue by diagnosis</i> , <i>Revenue trend</i>	Raw bill column, placeholders included. Report equivalent: validated revenue, 153,155 (Financial view, <i>Validated revenue</i> tile).
<i>Avg bill</i> tile (2,506)	Raw bill column divided by 247 episodes. Report equivalent: mean validated bill, 2,945 on 52 genuine bills.
<i>Billing quality</i> alert (100 placeholder bills) and <i>Clinical correlations</i> cards	Only 999 treated as a placeholder; the 95 bills at 3,852 are kept. Report equivalent: 195 placeholder bills; stay-bill Pearson $r = 0.66$ on 50 genuine bills (the card shows 0.31 on 133 episodes).

All other dashboard values (volumes, case mix, severity, cover, length of stay, data quality counts) are identical to the report. The captures below were taken with every filter cleared.

9.2 Overview

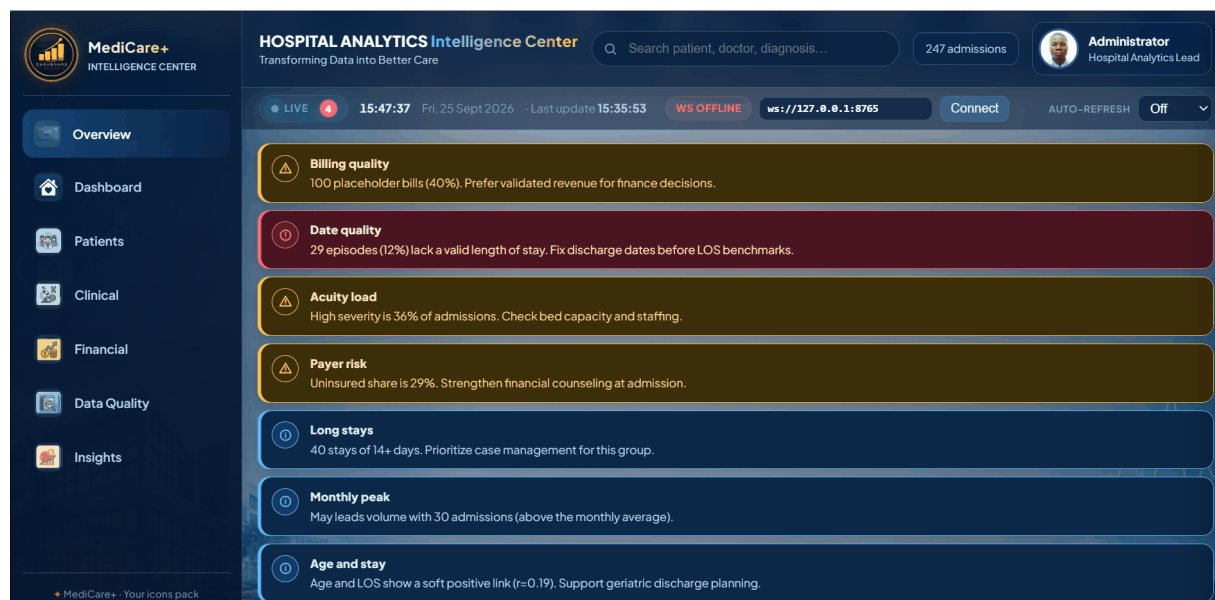


Figure 13: Overview, alert bar. Real-time alerts on billing quality, date quality, acuity, payer risk, long stays, monthly peak and age.



Figure 14: Overview, headline indicators with admissions by month, top diagnoses and top physicians.

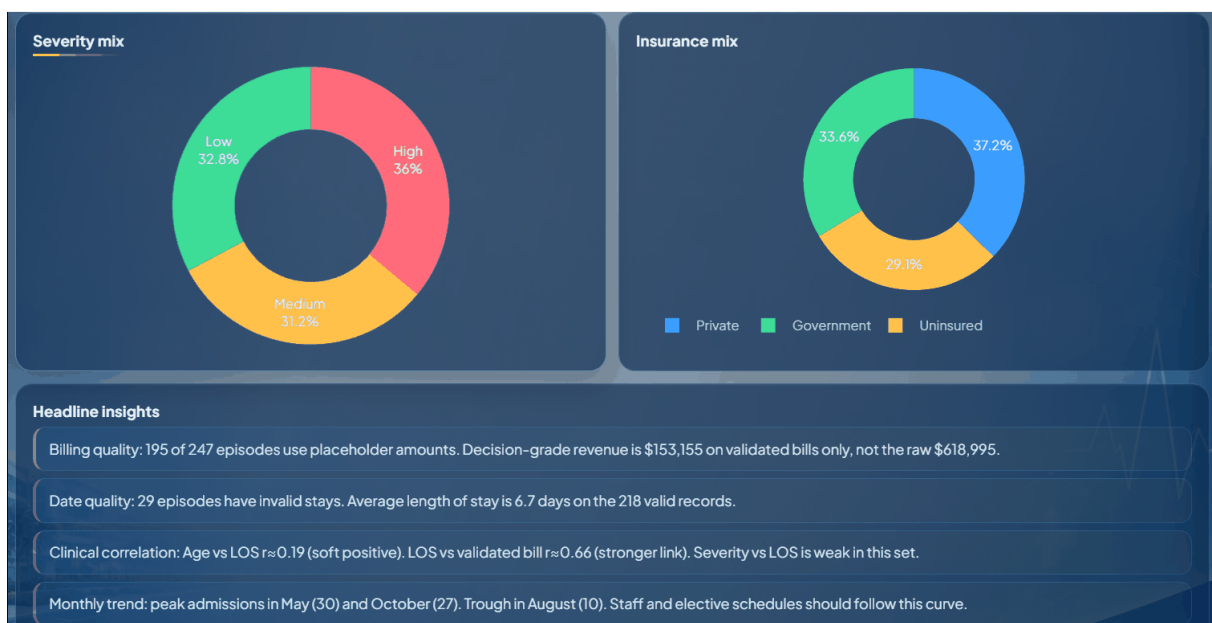


Figure 15: Overview, severity and insurance mix with the headline insights.

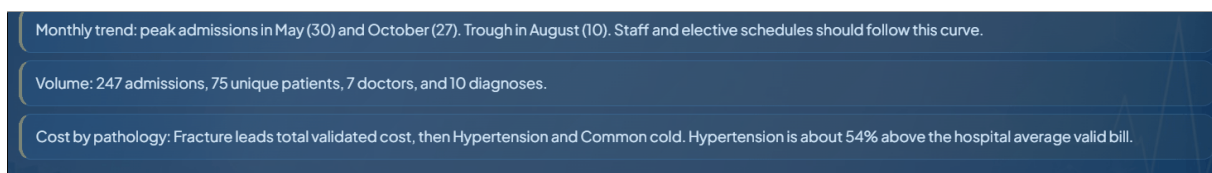


Figure 16: Overview, end of the headline insights: volume and cost by pathology.

9.3 Dashboard



Figure 17: Dashboard, indicator strip and filter bar (month, doctor, diagnosis, insurance, gender, severity).

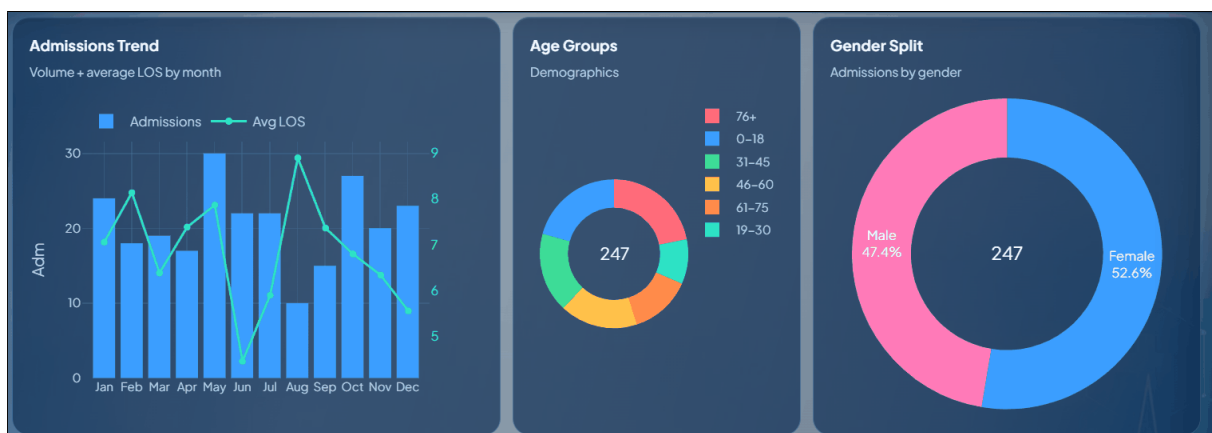


Figure 18: Dashboard, admissions and mean length of stay by month, age groups and gender split.

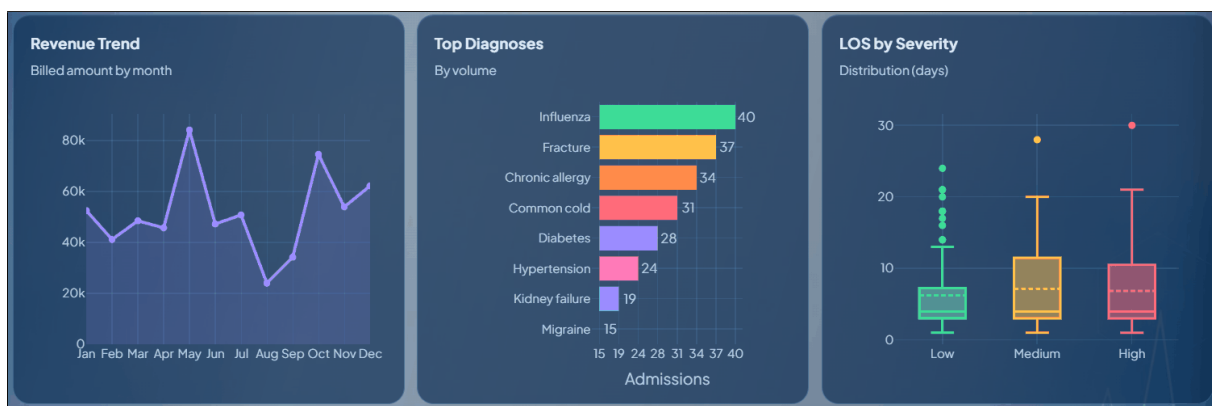


Figure 19: Dashboard, billed amount by month, top diagnoses and length of stay by severity.

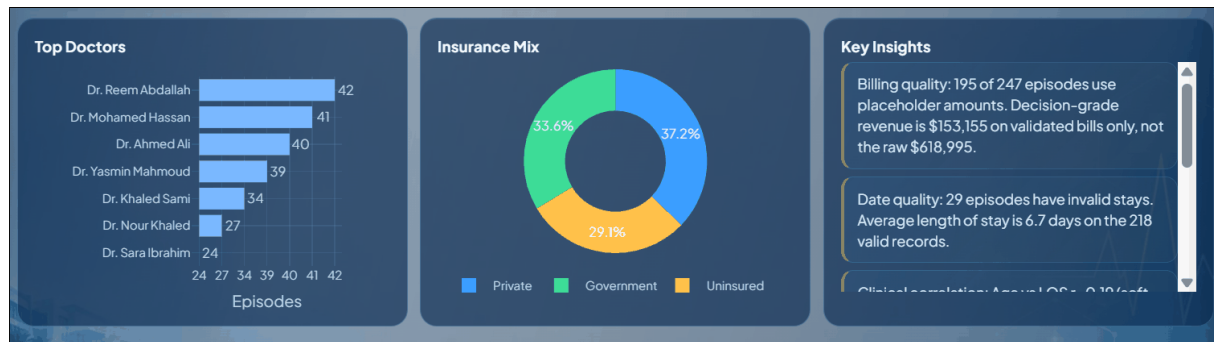


Figure 20: Dashboard, top physicians, insurance mix and key insights.

9.4 Patients

EPISODE	PATIENT	AGE	GENDER	DIAGNOSIS	SEVERITY	DOCTOR	ADMIT	LOS	BILL	INSURANCE
EP-0001	Mariam Khaled	22	Male	Fracture	Low	Dr. Yasmin Mahmoud	2023-01-01	4	\$5,339	Government
EP-0002	Mariam Youssef	57	Female	Chronic allergy	Low	Dr. Yasmin Mahmoud	2023-01-03	—	\$999	Private
EP-0003	Ibrahim Hassan	56	Male	Migraine	Low	Dr. Yasmin Mahmoud	2023-01-05	6	\$3,852	Uninsured
EP-0004	Mariam Youssef	84	Female	Diabetes	Low	Dr. Yasmin Mahmoud	2023-01-05	5	\$3,920	Government
EP-0005	Yasmin Hassan	26	Male	Kidney failure	High	Dr. Ahmed Ali	2023-01-05	2	\$999	Government
EP-0006	Ahmed Hassan	13	Male	Chronic allergy	Low	Dr. Mohamed Hassan	2023-01-06	4	\$999	Private
EP-0007	Mahmoud Abdallah	79	Female	Chronic allergy	High	Dr. Ahmed Ali	2023-01-06	4	\$999	Government
EP-0008	Ahmed El-Sayed	56	Female	Hypertension	Medium	Dr. Ahmed Ali	2023-01-08	20	\$999	Private
EP-0009	Ibrahim Mahmoud	60	Female	Influenza	Medium	Dr. Yasmin Mahmoud	2023-01-08	3	\$999	Uninsured
EP-0010	Sara Abdallah	38	Male	Influenza	Low	Dr. Sara Ibrahim	2023-01-10	3	\$999	Government
EP-0011	Sara Abdallah	89	Male	Stroke	High	Dr. Reem Abdallah	2023-01-10	13	\$3,852	Government
EP-0012	Sara Youssef	59	Female	Kidney failure	Medium	Dr. Yasmin Mahmoud	2023-01-14	7	\$999	Private
EP-0013	Mahmoud El-Sayed	81	Female	Common cold	High	Dr. Mohamed Hassan	2023-01-15	3	\$2,873	Uninsured
EP-0014	Mariam Hassan	70	Male	Hypertension	Low	Dr. Sara Ibrahim	2023-01-15	2	\$999	Private

Figure 21: Patients, the patient explorer: one row per episode with diagnosis, severity, physician, dates, stay, bill and cover.

EP-0015	Nada Ibrahim	41	Female	Hypertension	Low	Dr. Nour Khaled	2023-01-16	6	\$3,293	Private
EP-0016	Mahmoud Abdallah	65	Male	Hypertension	Low	Dr. Mohamed Hassan	2023-01-17	2	\$999	Uninsured
EP-0017	Mahmoud Mahmoud	70	Female	Diabetes	Medium	Dr. Reem Abdallah	2023-01-18	—	\$999	Private
EP-0018	Ibrahim Youssef	5	Female	Hypertension	Low	Dr. Yasmin Mahmoud	2023-01-19	13	\$3,852	Government
EP-0019	Mahmoud Khaled	11	Male	Influenza	High	Dr. Mohamed Hassan	2023-01-22	13	\$3,852	Uninsured
EP-0020	Ahmed Youssef	48	Male	Hypertension	Low	Dr. Reem Abdallah	2023-01-23	—	\$999	Uninsured
EP-0021	Mariam Ali	56	Female	Diabetes	Medium	Dr. Ahmed Ali	2023-01-23	5	\$999	Private
EP-0022	Ali Ibrahim	49	Female	Kidney failure	Low	Dr. Sara Ibrahim	2023-01-26	12	\$999	Uninsured
EP-0023	Nada Hassan	5	Male	Kidney failure	Low	Dr. Khaled Sami	2023-01-26	5	\$3,852	Government

← Prev Page 1/10 : 247 rows Next →

Figure 22: Patients, end of the first page of the explorer, with pagination over the 247 episodes.

9.5 Clinical

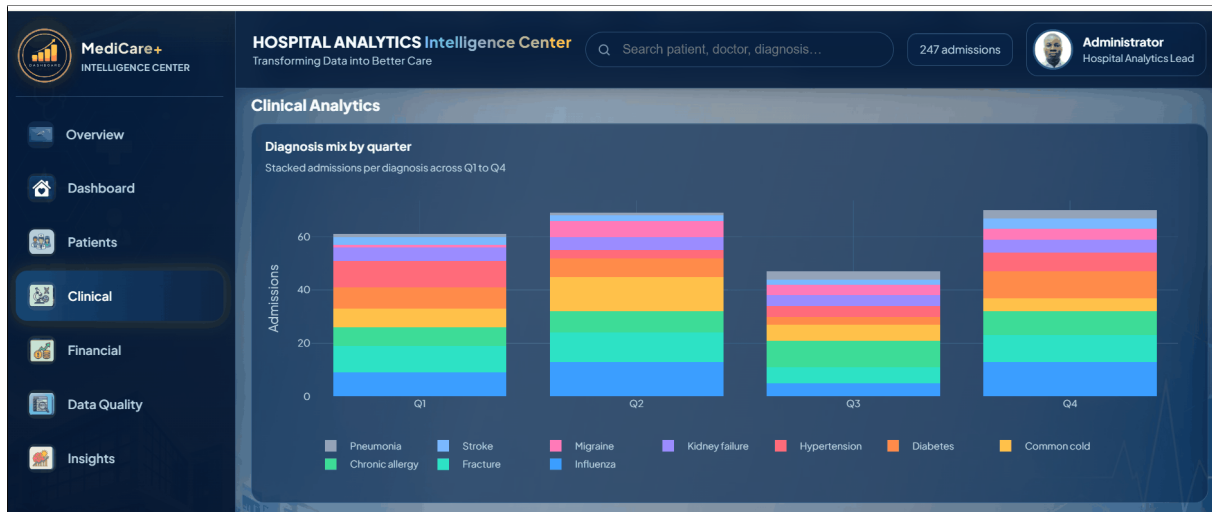


Figure 23: Clinical, diagnosis mix by quarter.

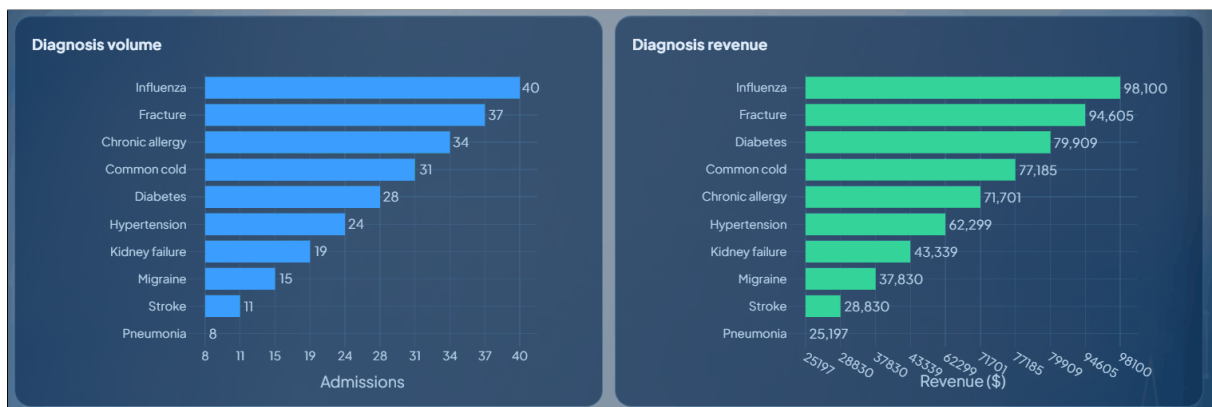


Figure 24: Clinical, diagnosis volume and billed amount by diagnosis.

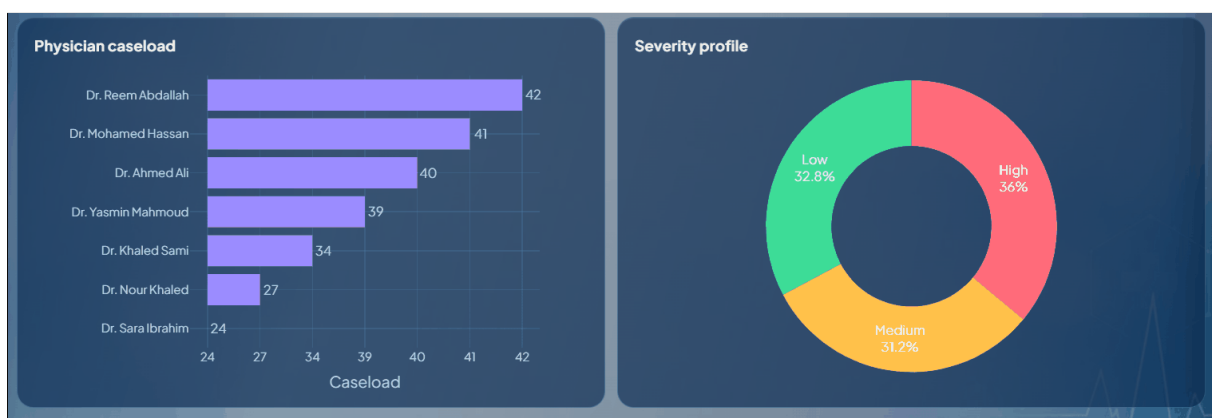


Figure 25: Clinical, physician caseload and severity profile.

9.6 Financial



Figure 26: Financial, total billed, validated revenue and average bill, with billed amount by insurance and by diagnosis.

9.7 Data Quality

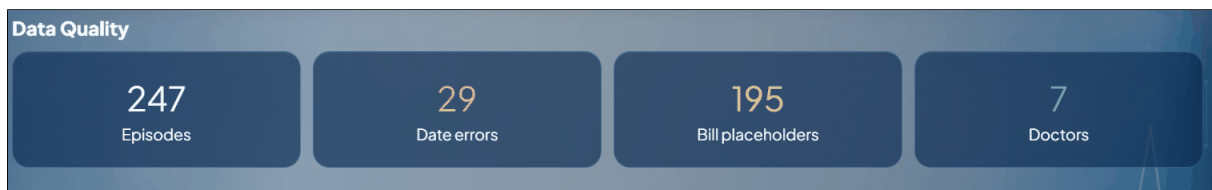


Figure 27: Data Quality, the defect counts: 247 episodes, 29 date errors, 195 placeholder bills, 7 physicians.

9.8 Insights

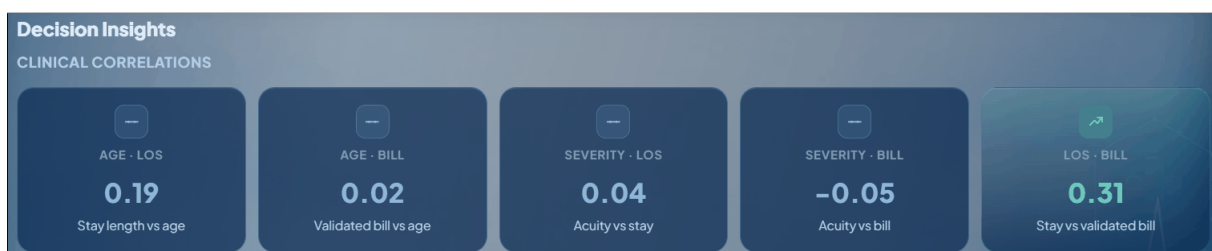


Figure 28: Insights, clinical correlation cards.



Figure 29: Insights, monthly snapshot of admissions, validated revenue and mean length of stay.

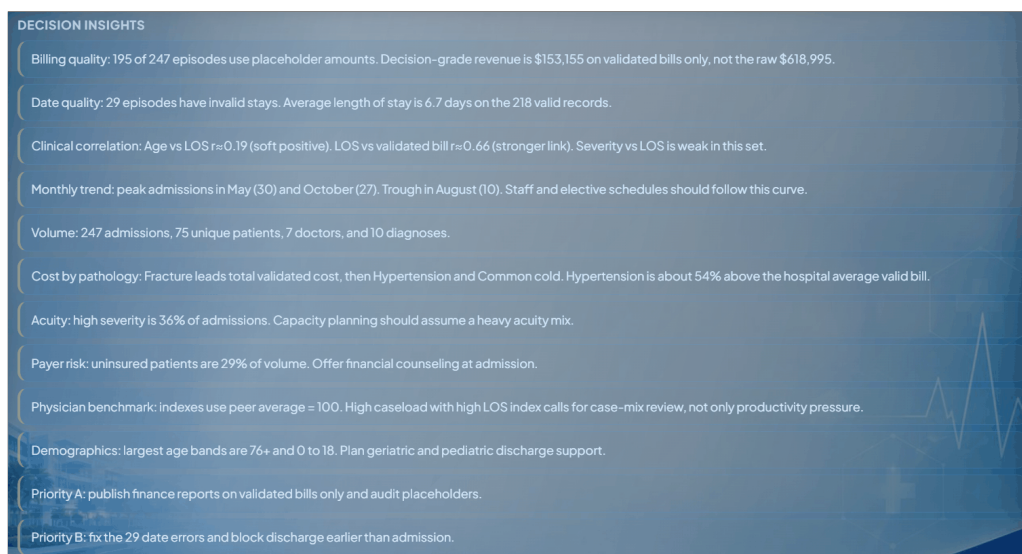


Figure 30: Insights, the decision insights.



Figure 31: Insights, priorities and the live-data dock for reloading a cleaned file.

10 Limitations

Revenue rests on 52 episodes, concentrated on short stays. No financial figure in this report is a budget input until the billing column is repaired at source. Means by diagnosis rest on 1 to 13 bills, and the stay–bill slope is only observed between 1 and 12 days.

Length of stay rests on 218 episodes, 88.3% of the file.

The sample is small for the number of segments. 247 episodes across 10 diagnoses and 7 physicians leaves several segments in single figures. Every segment comparison is reported with its test.

One facility, one year. There is no ward or site column, so no comparison between units is possible, and a single year supports no trend or seasonality claim.

Association, not causation. The stay–bill relationship shows that the bill follows the stay; it does not show why stays are long.

11 Reproducibility

Every number, table and figure of this report is produced by the notebook `MediCare-Hospital-Analytics.ipynb` from the raw file `data/hospital_raw.csv`. The notebook writes the clean register (`data/hospital_clean.csv`), the result tables (`results/`) and the figures (`figures/`, PNG at 300 dpi). Once the billing and date capture are repaired at source, re-running the notebook unchanged recomputes every flag, indicator, test and figure.

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